DSA0308-NATURAL LANGUAGE PROCESSING

ASSIGNMENT 5

CO5: LANGUAGE GENERATION AND DISCOURSE ANALYSIS

COURSE: NATURAL LANGUAGE PROCESSING

NATURE: INDIVIDUAL ASSIGNMENT

TOTAL MARKS: 50

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CONTENTS AND SUBMISSION NOTE

This report answers the tasks in the corresponding uploaded assignment. It follows the task order and preserves the assignment's terminology. The worked examples explain the reasoning and the evidence expected.

Important: where the assignment requires a corpus experiment, annotation, code execution or human judgement, no unrun result is presented as an actual measurement. Such values must be collected by running the experiment and inserted into the report. Any sample calculation is explanatory only.

A1 Discourse Analyst's World

Concepts and direct answer

Separate discourse analysis from generation. The analyser identifies discourse units, mentions, referents, relations and coreference links; the generator plans propositions and realises sentences. Define markables and measure precision=C/P, recall=C/R and F1=2PR/(P+R). Use the supplied gold links in Text T. Discuss ambiguity, topic shifts, quotation and uncertain references.

The answer begins by identifying the input, the target interpretation and the assumptions. Terms are defined before they are used. This prevents the same label from being applied to different meanings in different parts of the analysis.

The main conclusion is that the system must make its intermediate decisions explicit. A final label or output alone is not enough to show that the method is correct.

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Input	Sentence, corpus, query or scenario
Method	Formula, grammar, rule or algorithm
Trace	Intermediate steps and selected output
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A1 Discourse Analyst's World

Representation and method

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A1 Discourse Analyst's World

Worked example / trace

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Evaluation, limitations and conclusion

Separate discourse analysis from generation. The analyser identifies discourse units, mentions, referents, relations and coreference links; the generator plans propositions and realises sentences. Define markables and measure precision=C/P, recall=C/R and F1=2PR/(P+R). Use the supplied gold links in Text T. Discuss ambiguity, topic shifts, quotation and uncertain references.

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Conclusion: select a design only when its measured behaviour meets the task requirements. Discuss remaining limitations and the conditions under which a simpler alternative may be preferable.

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A2 Discourse Units and Graph

Concepts and direct answer

Segment Text T into coherent units using paragraph, topic and discourse cues. Build a discourse graph with nodes and labelled relations such as elaboration, sequence, cause, contrast and result. Attach new units on the right frontier. Compute next-mention information backward and state whether availability means future occurrence or right-frontier accessibility.

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A3 Generator Focus and Aggregation

Concepts and direct answer

Maintain a focus list of size k, entity realisation descriptors, aggregation buffer and given/new flag. Choose pronouns only when the antecedent is salient and unambiguous; use definite descriptions for known entities and indefinite NPs for new entities. Evict least salient entities when focus is full. Trace F1-F4 and compare word costs with naive full-name repetition.

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A4 Entity-Discourse Graph

Concepts and direct answer

The weak algorithm mistakes repeated nouns for coreference and misses implicit relations. Create one node per referent and mention links to nodes. Distinguish a new entity from a repeated reference using definiteness, identity evidence, semantic type and context. Attach relations using connective and semantic evidence. Demonstrate same noun/different entity, cataphor, one-anaphora and implicit causality.

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A5 Global Coherence and Salience

Concepts and direct answer

Explain coreference chains, lexical chains, centering, entity grids, discourse extraction and distributional similarity, with safety conditions. Define GEN and KILL sets and forward fixed-point equations IN=union predecessor OUT; OUT=GEN union (IN-KILL). State initialisation, join, boundary and termination. Compare backward presupposition analysis.

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B1 Instrumented Discourse Analyser

Concepts and direct answer

Implement segmentation, mention detection, graph construction, resolution and realisation with configurable focus size k. Record boundaries, candidate-set sizes, resolved/unresolved/ambiguous links, P/R/F1, emitted words and processed tokens. Test topic shift, same-head-noun distinct entities and a no-pronoun boundary case. Compare recency-only and combined salience rules for k=1,2,3,unlimited.

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B2 Adversarial Inputs

Concepts and direct answer

Construct two six-sentence texts with four pronouns: one easy and one designed to cause competing antecedents. Predict the first likely error before execution. Measure accuracy, unresolved/ambiguous links, candidate size and link changes after reordering. Explain why equal sentence/pronoun counts do not imply equal difficulty.

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B3 Dialogue Agent

Concepts and direct answer

Classify dialog acts such as inform, request, question, clarify, repair and thank. Track dialogue state from full history and from a bounded salient stack. Test information-seeking, clarification and repair dialogues. Include a conservative case where a hypothetical is not established and an unsafe case where removing clarification changes commitment. Measure operations as turns and entities increase.

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B4 Translation and Production Decision

Concepts and direct answer

Define none, fast and full processing levels combining reference resolution, discourse planning, realisation and translation. Resolve references before translation and plan before realisation. Report before/after pronouns, word/sentence counts, BLEU or chrF, human adequacy and operation counts. Include a case where full processing harms fluency or honorific/gender agreement. Recommend default level and language-independent interlingua or transfer separation.

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Evaluation, limitations and conclusion

Define none, fast and full processing levels combining reference resolution, discourse planning, realisation and translation. Resolve references before translation and plan before realisation. Report before/after pronouns, word/sentence counts, BLEU or chrF, human adequacy and operation counts. Include a case where full processing harms fluency or honorific/gender agreement. Recommend default level and language-independent interlingua or transfer separation.

Evaluation should include a normal case, a boundary case and an adversarial case. Report expected behaviour, actual behaviour, error category and a specific refinement. Use the same test inputs when comparing two designs.

Conclusion: select a design only when its measured behaviour meets the task requirements. Discuss remaining limitations and the conditions under which a simpler alternative may be preferable.

Evidence	Required content
Input	Sentence, corpus, query or scenario
Method	Formula, grammar, rule or algorithm
Trace	Intermediate steps and selected output
Validation	Metric, counterexample or comparison

Additional Analysis – Metric interpretation

Define every metric and denominator. Aggregate accuracy can hide poor performance on rare categories, unknown words or out-of-domain text.

Application to this assignment: CO5: LANGUAGE GENERATION AND DISCOURSE ANALYSIS. The design should preserve the input, expose intermediate decisions, state uncertainty and allow each stage to be checked independently.

Suggested evidence record: Input | Expected result | Actual result | Error category | Proposed correction. Complete this record using the student's own execution or annotation; do not present an illustrative value as a measured result.

Additional Analysis – Limitations

Separate data limitations from algorithm limitations. Explain which problem can be fixed with better examples and which requires a design change.

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Additional Analysis – Engineering trade-offs

Discuss accuracy, memory, latency, transparency, annotation cost and maintainability. A method is useful only relative to its operating constraints.

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Additional Analysis – Final recommendation

Choose the design that best satisfies the assignment's requirements and justify it using traceable evidence rather than a general claim that one method is better.

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Additional Analysis – Data and assumptions

State the data domain, language variety, preprocessing, annotation rules and excluded cases. Assumptions must be visible because they affect the interpretation of every result.

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Additional Analysis – Reproducibility

Record corpus source, software and library versions, random seed, parameters and train/test split. Save intermediate data and final output tables.

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Additional Analysis – Baseline comparison

A simple baseline provides a reference point. Compare every improvement with the same baseline, data and evaluation metric.

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Additional Analysis – Error analysis

Group errors by their cause, such as ambiguity, missing vocabulary, incorrect structure, weak context, annotation disagreement or domain shift.

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Additional Analysis – Boundary tests

Test empty input, one-token input, long input, unknown terms, punctuation variation and conflicting context. These cases expose hidden assumptions.

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Additional Analysis – Adversarial tests

Construct a difficult input that targets the method's weakest assumption. Predict the failure before testing and compare prediction with observation.

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CONCLUSION

The answers in this report follow the structure and concepts of the uploaded assignment. The central engineering lesson is to define representations and decision rules, trace the reasoning, test difficult cases, measure performance and defend the final design with evidence.

REFERENCES AND ACKNOWLEDGEMENT

Primary source: the corresponding DSA03 Natural Language Processing assignment PDF supplied for this task. Add exact textbooks, papers, corpora, toolkits and software references actually consulted, including author or organisation, year, title and publication details.

Disclose AI or reference tools used as required by the assignment. Verify the examples and replace all illustrative experimental values with results from your own implementation, corpus analysis or annotation.